Quiz Week 6

Groups

Groups and subgroups

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1. **Let a and b elements in a group G such that they commute, i.e., $ab = ba$. Mark the correct answer(s):**
   (a) a and b^2 commute.
   (b) a^{-1} and b commute.

Solution: Both (a) and (b) are correct.

(a) Multiply $ab = ba$ on the left by b to get $b(ab) = b^2a$. From here we have that

$$(ba)b = b(ab) = b^2a \overset{ab=ba}{\Longrightarrow} ab^2 = b^2a,$$

which shows that a and b commute.

(b) Multiplying $ab = ba$ on the right by a^{-1} gives $(ab)a^{-1} = (ba)a^{-1}$, that is, $aba^{-1} = b$. Multiplying again by a^{-1} but this time on the left gives $ba^{-1} = a^{-1}b$.

2. **In each case, indicate whether the statement is true or false.**
   (2) If H and K are subgroups of a group G, then $H \cap K$ is also a subgroup of G.
   (3) If H and K are subgroups of a group G, then $H \cup K$ is also a subgroup of G.
   (4) Every group contains at least two subgroups.
   (5) $H = \left\{ \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} : a,\, b \in \mathbb{R},\, a \neq 0 \right\}$ is a subgroup of $G = (GL_2(\mathbb{R}), \cdot)$.
   (6) $\mathbb{Z}_3$ is a subgroup of $\mathbb{Z}_6$.

Solution:

(2) **True:** We apply the Subgroup Test. Let us start by noticing that $1 \in H \cap K$ since $1 \in H$ and $1 \in K$ because they are subgroups of G. Next take $x,\, y \in K \cap H$, then we have that

$$xy \in K, \quad xy \in H, \quad x^{-1} \in K, \quad x^{-1} \in H,$$

which implies that

$$xy \in K \cap H \quad \text{and} \quad x^{-1} \in K \cap H,$$

concluding the proof.

(3) **False:** Take $G = \mathbb{Z}$, $H = 2\mathbb{Z}$ and $K = 3\mathbb{Z}$. Then $H \cup K$ consists on the integers that are either a multiple of 2, or a multiple of 3. Notice that both 2 and 3 are in $H \cup K$ but $2 + 3 = 5$ is not in $H \cup K$.

(4) **True:** Any group G has two subgroups: $\{1\}$ and G.

(5) **True:** Notice that $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in H$. Next, if $\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}$ and $\begin{pmatrix} c & d \\ 0 & c \end{pmatrix}$ are in H, then

$$\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}\begin{pmatrix} c & d \\ 0 & c \end{pmatrix} = \begin{pmatrix} ac & ad+bc \\ 0 & ac \end{pmatrix} \in H$$

since $ac \neq 0$ because a and c are nonzero real numbers. To finish, note that

$$\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}^{-1} = \begin{pmatrix} a^{-1} & -a^{-1}ba^{-1} \\ 0 & a^{-1} \end{pmatrix} \in H,$$

since $-a^{-1} \neq 0$.

(6) **False:** $\mathbb{Z}_3$ is not a subset of $\mathbb{Z}_6$. To avoid confusion we use the notation $[a]_n$ to denote $\bar{a} \in \mathbb{Z}_n$. Recall that

$$\mathbb{Z}_3 = \{[0]_3,\ [1]_3,\ [2]_3\},$$
$$\mathbb{Z}_6 = \{[0]_6,\ [1]_6,\ [2]_6,\ [3]_6,\ [4]_6,\ [5]_6\}$$

Notice that using the bar notation, namely, $\bar{a}$ will lead you to the wrong answer. Now, notice that none of the elements of $\mathbb{Z}_3$ belongs to $\mathbb{Z}_6$; for instance, $[0]_3$ does not coincide with any of the elements of $\mathbb{Z}_6$ since it consists on the integers that are multiples of 3 (the remainder when divided by 3 is 0).